

Discrete Mathematics (SCS 1302)

Short Note

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1 Propositional Logic

1.1 Propositions and Connectives

- **Proposition:** a declarative sentence that is either true or false, not both.
- **Atomic proposition:** cannot be broken into smaller statements. Denoted by variables $p, q, r, s, \dots$
- **Compound proposition:** formed by combining atomic propositions with **logical operators (connectives)**.

Operator	Symbol	Truth condition
Negation	$\neg p$	true when p is false
Conjunction (AND)	$p \wedge q$	true when both p, q true
Disjunction (OR)	$p \vee q$	false only when both p, q false
Exclusive OR	$p \oplus q$	true when exactly one of p, q true
Conditional (implication)	$p \rightarrow q$	false only when p true, q false
Biconditional	$p \leftrightarrow q$	true when p, q have same truth value

Conditional $p \rightarrow q$: p = hypothesis/antecedent, q = conclusion/consequent. Read as “ p is sufficient for q ”, “ q is necessary for p ”, “ p only if q ”, “ q whenever p ”.

- **Converse:** $q \rightarrow p$
- **Inverse:** $\neg p \rightarrow \neg q$
- **Contrapositive:** $\neg q \rightarrow \neg p$ ($p \rightarrow q \equiv \neg q \rightarrow \neg p$, same truth value)

Operator precedence (highest to lowest): $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$.

1.2 Truth Tables

For n atomic propositions, a truth table has 2^n rows (list all combinations, then evaluate sub-expressions in precedence order).

- **Tautology:** always true. **Contradiction:** always false. **Contingency:** neither (mix of T/F).
- **Valid:** a proposition is valid if its truth table column is all T.
- **Satisfiable:** at least one T in the truth table (tautology or contingency). **Unsatisfiable:** all F (contradiction).
- **Logically equivalent** ($P \equiv Q$): same truth table; equivalently $P \leftrightarrow Q$ is a tautology.
- **Logical consequence:** B is a logical consequence of $A_1, \dots, A_n$ if $(A_1 \wedge \dots \wedge A_n) \rightarrow B$ is valid (a tautology).
- **Consistency:** a set of propositions is consistent iff there is an assignment of truth values making all of them true simultaneously (e.g. system specifications should be consistent – no conflicting requirements).

1.3 Logical Equivalences

Identity	Name
$p \wedge T \equiv p, p \vee F \equiv p$	Identity laws
$p \vee T \equiv T, p \wedge F \equiv F$	Domination laws
$p \vee p \equiv p, p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation
$p \vee q \equiv q \vee p, p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	Distributive laws
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$	De Morgan's law
$\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's law
$p \vee (p \wedge q) \equiv p, p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv T, p \wedge \neg p \equiv F$	Complement laws

Conditional/biconditional equivalences:

$$\begin{aligned}
 p \rightarrow q &\equiv \neg p \vee q & p \rightarrow q &\equiv \neg q \rightarrow \neg p & \neg(p \rightarrow q) &\equiv p \wedge \neg q \\
 (p \rightarrow q) \wedge (p \rightarrow r) &\equiv p \rightarrow (q \wedge r) & (p \rightarrow r) \wedge (q \rightarrow r) &\equiv (p \vee q) \rightarrow r \\
 p \leftrightarrow q &\equiv (p \rightarrow q) \wedge (q \rightarrow p) & p \leftrightarrow q &\equiv (p \wedge q) \vee (\neg p \wedge \neg q) & \neg(p \leftrightarrow q) &\equiv p \leftrightarrow \neg q
 \end{aligned}$$

Principle of Duality: swap $\wedge \leftrightarrow \vee$ and $T \leftrightarrow F$; if a proposition is a tautology, its dual is also a tautology.

Functionally complete sets: $\{\neg, \wedge, \vee\}$; also $\{\neg, \wedge\}$, $\{\neg, \vee\}$, $\{\uparrow\}$ (NAND alone), $\{\downarrow\}$ (NOR alone).

1.4 Normal Forms

- **Literal:** a variable or its negation.
- **Elementary product:** conjunction ($\wedge$) of literals, e.g. $p \wedge q \wedge \neg r$.
- **Elementary sum:** disjunction ($\vee$) of literals, e.g. $p \vee \neg q \vee r$.
- **Disjunctive Normal Form (DNF):** a sum ($\vee$) of elementary products.
- **Conjunctive Normal Form (CNF):** a product ($\wedge$) of elementary sums.

Method 1 (algebraic manipulation): use equivalences to convert the proposition step by step, e.g. eliminate $\rightarrow, \leftrightarrow$ first via $p \rightarrow q \equiv \neg p \vee q$, then push negations inward with De Morgan's, then apply distributive laws.

Example (DNF): $(\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q)$

$$\begin{aligned}
 &\equiv \neg(\neg p \vee \neg q) \vee (p \leftrightarrow \neg q) \\
 &\equiv (p \wedge q) \vee [(p \wedge \neg q) \vee (\neg p \wedge q)] \\
 &\equiv (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q) \quad (\text{DNF})
 \end{aligned}$$

Example (CNF), same proposition:

$$\begin{aligned}
&\equiv (p \wedge q) \vee [(p \rightarrow \neg q) \wedge (\neg q \rightarrow p)] \\
&\equiv (p \wedge q) \vee [(\neg p \vee \neg q) \wedge (q \vee p)] \\
&\equiv [p \vee ((\neg p \vee \neg q) \wedge (q \vee p))] \wedge [q \vee ((\neg p \vee \neg q) \wedge (q \vee p))] \\
&\equiv [(p \vee \neg p \vee \neg q) \wedge (p \vee q \vee p)] \wedge [(q \vee \neg p \vee \neg q) \wedge (q \vee q \vee p)] \\
&\equiv [T \wedge (p \vee q)] \wedge [T \wedge (p \vee q)] \equiv (p \vee q) \quad (\text{CNF})
\end{aligned}$$

Method 2 (truth table): build the truth table; **DNF** = OR of one elementary product per row where output is T (variable un-negated if 1, negated if 0). **CNF** = AND of one elementary sum per row where output is F (variable negated if 1, un-negated if 0).

1.5 Applications

- **Translating English $\rightarrow$ logic:** identify atomic propositions, assign variables, combine with connectives. Note: “ p only if q ” $\equiv p \rightarrow q$.
- **System specifications:** check consistency using a truth table across all specifications simultaneously – consistent iff some row makes all specifications true.

2 Predicate Logic

2.1 Predicates and Quantifiers

- **Predicate** (propositional function) $P(x_1, \dots, x_n)$: a statement whose truth value depends on variable(s); becomes a proposition once variables are instantiated (given values).
- **Truth set** of $P(x)$ over domain U : $\{x \in U : P(x) \text{ is true}\}$.
- **Universal quantifier** $\forall x P(x)$: “ $P(x)$ true for every x in domain”. False iff there is an x for which $P(x)$ is false (a counterexample).
- **Existential quantifier** $\exists x P(x)$: “there is at least one x with $P(x)$ true”. False iff $P(x)$ is false for every x .
- If domain is empty: $\forall x P(x)$ is (vacuously) true; $\exists x P(x)$ is false.
- Quantifiers have higher precedence than logical operators: $\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$, not $\forall x (P(x) \vee Q(x))$.

2.2 Negation of Quantifiers (De Morgan’s for quantifiers)

$$\neg \forall x P(x) \equiv \exists x \neg P(x) \qquad \neg \exists x Q(x) \equiv \forall x \neg Q(x)$$

To negate a statement so no negation precedes a quantifier: push $\neg$ inward, flipping $\forall \leftrightarrow \exists$ each time it crosses a quantifier. *Example:* $\neg(\forall x \exists y (xy = 1)) \equiv \exists x \neg(\exists y (xy = 1)) \equiv \exists x \forall y \neg(xy = 1) \equiv \exists x \forall y (xy \neq 1)$.

2.3 Logical Equivalences Involving Quantifiers

$$\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$$

$$\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$$

(Note: $\forall x (P(x) \vee Q(x)) \not\equiv \forall x P(x) \vee \forall x Q(x)$ in general, and similarly $\exists x (P(x) \wedge Q(x)) \not\equiv \exists x P(x) \wedge \exists x Q(x)$ in general.)

2.4 Binding and Scope

A variable bound by a quantifier has no free (unbound) occurrence; an expression with a free variable is not a proposition (its truth value is undetermined) until that variable is also quantified or instantiated. Example: in $\forall x[\forall y xy = x \rightarrow x = 0]$, the inner $\forall y$ has scope only over $xy = x$; the outer $\forall x$ has scope over the whole bracket.

2.5 Nested Quantifiers

Order of quantifiers matters when quantifier types differ: $\forall x \exists y P(x, y) \not\equiv \exists y \forall x P(x, y)$ in general (same-type quantifiers, e.g. $\forall x \forall y$, can be swapped freely). Domain of x, y, z often $\mathbb{R}$ or $\mathbb{Z}$; e.g. commutative law $\forall x \forall y (x + y = y + x)$, existence of additive inverse $\forall x \exists y (x + y = 0)$.

2.6 Rules of Inference for Quantified Statements

Rule	Name
$\frac{\forall x P(x)}{\therefore P(c)}$	Universal Instantiation
$\frac{P(c) \text{ for arbitrary } c}{\therefore \forall x P(x)}$	Universal Generalization
$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$	Existential Instantiation
$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$	Existential Generalization

Note: in Existential Instantiation, c is a **new** witness – once chosen, the **same** c must be used consistently in every subsequent step that derives a conclusion about that particular object (cannot substitute a different element mid-proof).

2.7 Translating Sentences with Predicates

Example: domain = integers, $N(x)$: x non-negative, $E(x)$: x even, $O(x)$: x odd, $P(x)$: x prime.

- “There exists an odd prime”: $\exists x (O(x) \wedge P(x))$
- “All primes are non-negative”: $\forall x (P(x) \rightarrow N(x))$
- “Every integer is even or odd”: $\forall x (E(x) \vee O(x))$
- “The only even prime is 2”: $\forall x (P(x) \wedge E(x) \rightarrow x = 2)$
- “If not odd then even”: $\forall x (\neg O(x) \rightarrow E(x))$

3 Arguments (Rules of Inference)

3.1 Valid Arguments

- An **argument** is a sequence of propositions: all but the last are **premises**, the last is the **conclusion**.
- **Valid:** truth of all premises implies truth of conclusion, i.e. $(\text{premise}_1 \wedge \dots \wedge \text{premise}_n) \rightarrow \text{conclusion}$ is a tautology.
- Test validity: (a) truth table – check the implication above is always T; or (b) apply rules of inference step by step from premises to conclusion.

3.2 Rules of Inference for Propositional Logic

Rule	Name	Rule	Name
$\frac{p, p \rightarrow q}{\therefore q}$	Modus Ponens	$\frac{p}{\therefore p \vee q}$	Addition
$\frac{\neg q, p \rightarrow q}{\therefore \neg p}$	Modus Tollens	$\frac{p \wedge q}{\therefore p}$	Simplification
$\frac{p \rightarrow q, q \rightarrow r}{\therefore p \rightarrow r}$	Hypothetical Syllogism	$\frac{p, q}{\therefore p \wedge q}$	Conjunction
$\frac{p \vee q, \neg p}{\therefore q}$	Disjunctive Syllogism	$\frac{p \vee q, \neg p \vee r}{\therefore q \vee r}$	Resolution

Worked example: Premises: “If e-mail then finish program”, “If not e-mail then sleep early”, “If sleep early then wake refreshed”. Show: “If not finish program then wake refreshed”. Let p =e-mail, q =finish program, r =sleep early, s =wake refreshed.

Step	Statement	Reason
1	$p \rightarrow q$	Premise
2	$\neg q \rightarrow \neg p$	Contrapositive of (1)
3	$\neg p \rightarrow r$	Premise
4	$\neg q \rightarrow r$	Hyp. syllogism (2)(3)
5	$r \rightarrow s$	Premise
6	$\neg q \rightarrow s$	Hyp. syllogism (4)(5)

Fallacies (invalid forms) to avoid:

- **Affirming the consequent:** $p \rightarrow q, q \therefore p$ – **invalid**.
- **Denying the antecedent:** $p \rightarrow q, \neg p \therefore \neg q$ – **invalid**.

3.3 Rules of Inference for Quantified Statements

See Section 2.6 (Universal/Existential Instantiation/Generalization). Combined with propositional rules of inference to build multi-step proofs about predicates.

Worked example: Premises: “A student in this class has not read the book” ($\exists x(C(x) \wedge \neg B(x))$), “Everyone in this class passed the exam” ($\forall x(C(x) \rightarrow P(x))$). Conclude: “Someone who passed has not read the book” ($\exists x(P(x) \wedge \neg B(x))$).

1	$\exists x(C(x) \wedge \neg B(x))$	Premise
2	$C(a) \wedge \neg B(a)$	Existential instantiation (1)
3	$C(a)$	Simplification (2)
4	$\forall x(C(x) \rightarrow P(x))$	Premise
5	$C(a) \rightarrow P(a)$	Universal instantiation (4)
6	$P(a)$	Modus ponens (3)(5)
7	$\neg B(a)$	Simplification (2)
8	$P(a) \wedge \neg B(a)$	Conjunction (6)(7)
9	$\exists x(P(x) \wedge \neg B(x))$	Existential generalization (8)

Checking invalidity with a countermodel: to show an argument with quantifiers is **not** valid, exhibit a domain and an interpretation (e.g. a small Venn diagram / finite model) under which all premises are true but the conclusion is false. *Example:* “Some scientists are not engineers”, “Some astronauts are not engineers” $\therefore$ “Some scientists are not astronauts” – **not valid** (construct sets S, A, E overlapping so premises hold but every non-engineer scientist happens to also be a non-engineer astronaut, i.e. $S \subseteq A$ works as a countermodel).

3.4 Resolution Principle (used for resolution refutation)

- **Literal:** a variable or negation of a variable.
- **Clause:** a disjunction (sum) of literals.
- **Resolvent:** given clauses C_1, C_2 with complementary literals $p \in C_1, \neg p \in C_2$: delete $p, \neg p$ and take the disjunction of what remains. *Example:* $C_1 = p \vee q \vee \neg s, C_2 = \neg p \vee u \Rightarrow$ resolvent $= q \vee \neg s \vee u$.
- **Theorem:** a resolvent C of C_1, C_2 is a logical consequence of C_1, C_2 . (Modus Ponens is the special case: $p, \neg p \vee q \Rightarrow$ resolvent q .)
- **Resolution (reduction) of C from a clause set S :** a finite sequence $C_1, \dots, C_k$ where each C_i is either in S or a resolvent of two earlier clauses in the sequence, and $C_k = C$.
- **Refutation / proof of S :** a resolution deriving the **empty clause** $\square$ from S . If $\square$ is derived, S is **inconsistent (unsatisfiable)**.
- **Clausal form:** convert each premise to CNF, then list each conjunct as a separate clause (drop the $\wedge$'s between them).

Worked example (Modus Tollens via resolution): $\neg q, p \rightarrow q (\equiv \neg p \vee q) \therefore \neg p$. Resolve $\neg q$ with $\neg p \vee q$ on q : resolvent $= \neg p$. Resolve $\neg p$ with premise p (if present) $\rightarrow$ empty clause $\square \Rightarrow$ refutation.

Procedure for resolution refutation of a knowledge base $KB \models \alpha$:

1. Convert every premise in KB and the **negation** of the desired conclusion $\neg\alpha$ into clausal form.
2. Repeatedly resolve pairs of clauses (on complementary literals) and add resolvents to the clause set.
3. If the empty clause $\square$ is derived, $KB \wedge \neg\alpha$ is unsatisfiable, so $KB \models \alpha$ (the conclusion follows / knowledge base is consistent only without $\neg\alpha$).
4. If no more resolvents can be produced and $\square$ is never derived, $KB \wedge \neg\alpha$ is satisfiable – exhibit the satisfying assignment.

4 Introduction to Proofs

4.1 Terminology

- **Theorem:** a statement shown to be true. **Lemma:** a simple theorem used to help prove another theorem. **Corollary:** a proposition following directly from a proved theorem. **Conjecture:** a statement whose truth is unknown (becomes a theorem once proven).
- Most theorems have the form $p \rightarrow q$ or $\forall x(P(x) \rightarrow Q(x))$.

4.2 Methods of Proving $p \rightarrow q$

- **Vacuous proof:** show p is false (then $p \rightarrow q$ is automatically true).
- **Trivial proof:** show q is true (then $p \rightarrow q$ is automatically true).
- **Direct proof:** assume p true; use definitions/axioms/theorems and rules of inference to show q must be true.

- **Proof by contraposition** (indirect): since $p \rightarrow q \equiv \neg q \rightarrow \neg p$, assume $\neg q$ and show $\neg p$.
- **Proof by contradiction**: assume $\neg p$ (negation of what's to be proved) and derive a contradiction $r \wedge \neg r$; concludes original statement p true. To prove $p \rightarrow q$: assume $p \wedge \neg q$ and derive a contradiction.

4.2.1 Direct Proof – Worked Example

Theorem: The sum of two odd integers is even.

Proof: Let n, m be arbitrary odd integers, so $n = 2k_1 + 1$, $m = 2k_2 + 1$ for some $k_1, k_2 \in \mathbb{Z}$.

$$n + m = 2k_1 + 1 + 2k_2 + 1 = 2(k_1 + k_2 + 1) = 2k', \quad k' \in \mathbb{Z}$$

So $n + m$ is even. Since n, m arbitrary, sum of any two odd integers is even. ■

(Definition used: n is odd if $\exists k \in \mathbb{Z} : n = 2k + 1$; n is even if $\exists k \in \mathbb{Z} : n = 2k$.)

4.2.2 Proof by Contraposition – Worked Example

Theorem: For integer n , if n^2 is odd then n is odd.

Statement: $\forall n \in \mathbb{Z} (O(n^2) \rightarrow O(n))$, equivalent to $\forall n \in \mathbb{Z} (\neg O(n) \rightarrow \neg O(n^2))$, i.e. if n is even then n^2 is even.

Proof: Let n be arbitrary; suppose n even, so $n = 2k$.

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2) = 2k_1, \quad k_1 = 2k^2 \in \mathbb{Z}$$

So n^2 is even. This proves the contrapositive, hence the original statement.

4.2.3 Proof by Contradiction – Worked Example (irrationality)

Theorem: $\sqrt{2}$ is irrational.

Proof: Let p : “ $\sqrt{2}$ is irrational”. Suppose $\neg p$: $\sqrt{2}$ is rational.

$$\begin{aligned} \Rightarrow \sqrt{2} &= a/b, \quad b \neq 0, \quad \gcd(a, b) = 1 \\ \Rightarrow a^2 &= 2b^2 \Rightarrow a^2 \text{ even} \Rightarrow a \text{ even (since } a^2 \text{ even} \Rightarrow a \text{ even)} \\ \Rightarrow a &= 2k, \quad k \in \mathbb{Z} \Rightarrow 2b^2 = (2k)^2 = 4k^2 \Rightarrow b^2 = 2k^2 \Rightarrow b^2 \text{ even} \Rightarrow b \text{ even} \\ \Rightarrow \gcd(a, b) &\geq 2, \text{ contradicting } \gcd(a, b) = 1. \end{aligned}$$

Contradiction $\Rightarrow \neg p$ is false $\Rightarrow \sqrt{2}$ is irrational. ■

This is the exact template for proving $\sqrt{3}$ (or any $\sqrt{p}$, p prime) irrational: same structure, using “ $p \mid n^2 \Rightarrow p \mid n$ ” in place of the even/odd step.

4.3 Proofs of Equivalence

To prove $p \leftrightarrow q$: prove both $p \rightarrow q$ and $q \rightarrow p$.

To prove $p_1 \leftrightarrow p_2 \leftrightarrow \dots \leftrightarrow p_n$ all equivalent: prove a chain $p_1 \rightarrow p_2 \rightarrow \dots \rightarrow p_n \rightarrow p_1$ (circle of implications) – fewer proofs needed than all pairs.

4.4 Counterexample

To disprove $\forall x P(x)$, exhibit one specific x for which $P(x)$ is false.

4.5 Proof by Cases (Exhaustive Proof)

Split into an exhaustive set of cases covering every possibility; prove the result in each case separately. *Example:* show no integer solutions to $x^2 + 3y^2 = 8$ – bound $|x|, |y|$ using the equation, then check the finite remaining cases exhaustively.

4.6 Existence and Uniqueness Proofs

- **Existence proof** of $\exists x P(x)$: either **constructive** (exhibit a specific witness a with $P(a)$ true) or **non-constructive** (prove existence without exhibiting an explicit witness, e.g. via contradiction or a counting/pigeonhole argument).
- **Uniqueness proof**: show existence of x with $P(x)$, then show that if $P(y)$ also holds then $y = x$ (or assume two solutions $x_1 \neq x_2$ and derive a contradiction).

4.7 Backward Reasoning

Start from the desired conclusion q and work backwards, finding statements $p_1, p_2, \dots$ such that q follows from p_1 , p_1 follows from p_2 , etc., until reaching a known true statement – useful when a forward direct proof isn't obvious (e.g. inequality proofs).

5 Set Theory

5.1 Basics

- **Set**: a well-defined collection of distinct objects (elements/members); order doesn't matter, no duplicates.
- Notation: $x \in A$ (x member of A), $x \notin A$; $|A|$ = cardinality (number of elements).
- **Null/empty set** $\emptyset$ or $\{\}$: has no elements. Note $\{0\} \neq \emptyset$.
- **Universal set** $\mathbb{U}$: set of all elements under consideration.
- **Singleton set**: a set with exactly one element, e.g. $\{\emptyset\}$.
- **Equality**: $A = B$ iff they contain exactly the same elements.

5.2 Subsets and Power Set

- $A \subseteq B$ iff every element of A is also in B : $\forall x(x \in A \rightarrow x \in B)$.
- $A \subset B$ (proper subset): $A \subseteq B$ and $A \neq B$.
- $\emptyset \subseteq A$ for every set A (vacuously true); $A \subseteq A$ always.
- **Power set** $\mathcal{P}(A)$: the set of **all** subsets of A .
- **Theorem**: if $|A| = n$ then $|\mathcal{P}(A)| = 2^n$.
- **Counting subsets with constraints** (of an n -element set):
 - Subsets containing a specific element a : 2^{n-1} (the other $n-1$ elements each independently in/out).
 - Subsets containing two specific elements a, b : 2^{n-2} .
 - Subsets of exactly k elements: $\binom{n}{k}$.

- Subsets containing a but not e (2 fixed elements): 2^{n-2} (remaining $n - 2$ elements free).

5.3 Set Operations

- **Union:** $A \cup B = \{x : x \in A \vee x \in B\}$.
- **Intersection:** $A \cap B = \{x : x \in A \wedge x \in B\}$. **Disjoint** if $A \cap B = \emptyset$.
- **Difference:** $A - B = \{x : x \in A \wedge x \notin B\}$.
- **Complement:** $A^c = \overline{A} = \mathbb{U} - A$.
- **Symmetric difference:** $A \oplus B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.

5.4 Set Identities

Identity	Name
$A \cup \emptyset = A, A \cap \mathbb{U} = A$	Identity laws
$A \cup \mathbb{U} = \mathbb{U}, A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A, A \cap A = A$	Idempotent laws
$(A^c)^c = A$	Complementation law
$A \cup B = B \cup A, A \cap B = B \cap A$	Commutative laws
$(A \cup B) \cup C = A \cup (B \cup C)$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c$	De Morgan's laws
$A \cup (A \cap B) = A, A \cap (A \cup B) = A$	Absorption laws
$A \cup A^c = \mathbb{U}, A \cap A^c = \emptyset$	Complement laws

5.4.1 Methods for Proving Set Identities

1. **Element-membership (usual) method:** show $A \subseteq B$ and $B \subseteq A$ separately by taking arbitrary $x \in A$ and showing $x \in B$, and vice versa.
2. **Set-builder method:** rewrite both sides using set-builder notation and logical connectives, then apply logical equivalences.
3. **Membership table method:** like a truth table but with 1 (in the set) / 0 (not in the set) for each region; two expressions are equal iff their columns match for every combination.
4. **Using known set identities:** substitute step by step, naming each law used (analogous to algebraic proof of logical equivalences).

Worked example (element-membership): Prove $A - (B \cup C) = (A - B) \cap (A - C)$.

$$\begin{aligned}
x \in A - (B \cup C) &\iff x \in A \wedge x \notin (B \cup C) \\
&\iff x \in A \wedge \neg(x \in B \vee x \in C) \\
&\iff x \in A \wedge (x \notin B \wedge x \notin C) \\
&\iff (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C) \\
&\iff x \in (A - B) \wedge x \in (A - C) \iff x \in (A - B) \cap (A - C)
\end{aligned}$$

Since both directions hold for arbitrary x , the sets are equal. ■

Worked example (using identities): Show $(A \cup B) \cap (A \cup B^c) = A$.

$$\begin{aligned}(A \cup B) \cap (A \cup B^c) &\equiv A \cup (B \cap B^c) \quad (\text{Distributive law}) \\ &\equiv A \cup \emptyset \quad (\text{Complement law}) \\ &\equiv A \quad (\text{Identity law})\end{aligned}$$

5.5 Bit-String Representation of Sets

For finite universal set $\mathbb{U} = \{a_1, \dots, a_n\}$ (fixed order), any subset $A \subseteq \mathbb{U}$ is represented as an n -bit string: bit i is 1 if $a_i \in A$, else 0.

- $A \cup B$: **bitwise OR** of the two bit strings.
- $A \cap B$: **bitwise AND**.
- A^c : **bitwise complement** (flip every bit).
- $A - B = A \cap B^c$: AND of A 's string with the complement of B 's string.
- $A \oplus B$: **bitwise XOR**.

Example: $\mathbb{U} = \{1, \dots, 10\}$. Odd integers: 1010101010. Integers ≤ 5 : 1111100000. Then $(\text{odd}) \cap (\leq 5)$ via bitwise AND: 1010101010 AND 1111100000 = 1010100000 (i.e. $\{1, 3, 5\}$).

5.6 Generalized Union and Intersection

For sets $A_1, \dots, A_n$: $\bigcup_{i=1}^n A_i$ = elements in **at least one** A_i ; $\bigcap_{i=1}^n A_i$ = elements in **all** A_i .

5.7 Truth Sets and Quantifiers

Truth set of predicate $P(x)$ over domain D : $T_P = \{x \in D : P(x) \text{ true}\}$. $\forall x P(x)$ true over $\mathbb{U} \iff T_P = \mathbb{U}$; $\exists x P(x)$ true $\iff T_P \neq \emptyset$.

6 Functions

6.1 Basics

- A **function** $f : A \rightarrow B$ assigns to **each** element of A (domain) **exactly one** element of B (codomain). Written $f(a) = b$.
- **Range** of f : $\{f(a) : a \in A\} \subseteq B$ (may be a proper subset of codomain).
- Requirement: every domain element must be assigned **some** value, and **only one** value (else not a function).

6.2 One-to-One (Injective)

Definition: f is one-to-one iff $f(a) = f(b) \Rightarrow a = b$ for all a, b in domain.

$$f \text{ one-to-one} \iff \forall a \forall b (f(a) = f(b) \rightarrow a = b) \iff \forall a \forall b (a \neq b \rightarrow f(a) \neq f(b))$$

Check: no two distinct domain elements map to the same codomain element.

6.3 Onto (Surjective)

Definition: $f : A \rightarrow B$ is onto iff every element of B is the image of some element of A .

$$f \text{ onto} \iff \forall y \in B \exists x \in A : f(x) = y \iff \text{Range}(f) = B$$

6.4 Bijection (One-to-one Correspondence)

f is a **bijection** iff it is both one-to-one and onto. Bijections are exactly the functions that have an **inverse function**.

6.5 Inverse Function

Definition: if $f : A \rightarrow B$ is a bijection, $f^{-1} : B \rightarrow A$ assigns to each $b \in B$ the unique $a \in A$ with $f(a) = b$: $f^{-1}(b) = a \iff f(a) = b$. (Only bijections have inverses.)

6.6 Composition

Definition: for $g : A \rightarrow B$, $f : B \rightarrow C$: $(f \circ g)(a) = f(g(a))$ for all $a \in A$. Requires range of $g \subseteq \text{domain of } f$.

- If f, g both one-to-one, $f \circ g$ is one-to-one.
- If f, g both onto, $f \circ g$ is onto.
- If f, g both bijections, $f \circ g$ is a bijection and $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

6.7 Identity Function

$\iota_A : A \rightarrow A$, $\iota_A(x) = x$ for all x . For any bijection $f : A \rightarrow B$: $f^{-1} \circ f = \iota_A$, $f \circ f^{-1} = \iota_B$.

6.8 Image and Inverse Image of a Subset

For $f : A \rightarrow B$, $S \subseteq A$, $T \subseteq B$:

$$f(S) = \{f(s) : s \in S\} \qquad f^{-1}(T) = \{a \in A : f(a) \in T\}$$

6.9 Floor and Ceiling Functions

- **Floor** $\lfloor x \rfloor$: largest integer $\leq x$ (greatest integer function).
- **Ceiling** $\lceil x \rceil$: smallest integer $\geq x$.
- Examples: $\lfloor 0.5 \rfloor = 0$, $\lfloor 7.25 \rfloor = 7$, $\lfloor -12.4 \rfloor = -13$, $\lfloor -10 \rfloor = -10$; $\lceil 0.5 \rceil = 1$, $\lceil 7.25 \rceil = 8$, $\lceil -12.4 \rceil = -12$, $\lceil -10 \rceil = -10$.
- Note: for negative non-integers, floor rounds *away* from zero (more negative), ceiling rounds *toward* zero.

6.10 Partial Functions

A **partial function** f from A to B assigns to **some** (not necessarily all) elements of A a unique element of B ; if f is defined at a , f is **total** at a (a function in the usual sense is a partial function total at every domain element).

7 Binary Relations and Their Properties

7.1 Basics

- A **binary relation** R from A to B is a subset of $A \times B$: $R \subseteq A \times B$. Write aRb or $(a, b) \in R$.
- A relation **on** a set A : $R \subseteq A \times A$.

- **Counting:** since $|A \times A| = n^2$ for $|A| = n$, and the number of subsets of a set with m elements is 2^m , there are 2^{n^2} different relations on an n -element set.

7.2 Properties of Relations (on a set A)

Property	Definition
Reflexive	$\forall a [(a, a) \in R]$
Symmetric	$\forall a \forall b [(a, b) \in R \rightarrow (b, a) \in R]$
Antisymmetric	$\forall a \forall b [((a, b) \in R \wedge (b, a) \in R) \rightarrow a = b]$
Transitive	$\forall a \forall b \forall c [((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R]$

Note: symmetric and antisymmetric are **not** opposites – a relation can be both (e.g. equality) or neither.

7.3 Combining Relations

For relations R, S (subsets of $A \times B$): ordinary set operations apply directly: $R \cup S$, $R \cap S$, $R - S$, $R \oplus S$.

7.4 Composition of Relations

Definition: R relation from A to B , S relation from B to C . Composite $S \circ R$ (relation from A to C):

$$S \circ R = \{(a, c) \in A \times C : \exists b \in B [(a, b) \in R \wedge (b, c) \in S]\}$$

7.5 Powers of a Relation

Definition: R on set A . Powers defined recursively: $R^1 = R$, $R^{n+1} = R^n \circ R$. (So $R^2 = R \circ R$, $R^3 = R^2 \circ R = (R \circ R) \circ R$, etc.)

Worked example: $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$.

$$R^2 = \{(1, 1), (2, 1), (3, 1), (4, 2)\}$$

$$R^3 = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$$

$$R^4 = \{(1, 1), (2, 1), (3, 1), (4, 1)\} = R^3 \Rightarrow R^n = R^3, \forall n \geq 4$$

Theorem: R is transitive $\iff R^n \subseteq R$ for all $n \geq 1$.

7.6 Worked Example: Set of Students/Courses

$R_1 = \{(a, b) : a \text{ has taken course } b\}$, $R_2 = \{(a, b) : a \text{ needs } b \text{ to graduate}\}$.

- $R_1 \cup R_2$: has taken b **or** needs b .
- $R_1 \cap R_2$: has taken b **and** needs b (already satisfied requirement).
- $R_1 - R_2$: has taken b but doesn't need it.
- $R_2 - R_1$: needs b but hasn't taken it (still to complete).

8 Number Theory

8.1 Divisibility

- $a \mid b$ (" a divides b "): $a \neq 0$ and $\exists c \in \mathbb{Z} : b = ac$. a is a **factor/divisor** of b ; b is a **multiple** of a . Quantifier form: $\exists c(ac = b)$.

- **Theorem 1** (basic facts, $a \neq 0$): (1) $a \mid b \wedge a \mid c \Rightarrow a \mid (b + c)$; (2) $a \mid b \Rightarrow a \mid bx \ \forall x \in \mathbb{Z}$; (3) $a \mid b \wedge b \mid c \Rightarrow a \mid c$.

Proof of (1): $b = as, c = at$ for some $s, t \in \mathbb{Z} \Rightarrow b + c = a(s + t)$, so $a \mid (b + c)$.

8.2 The Division Algorithm

Theorem: for integer a , positive integer d , there are **unique** integers q, r with $0 \leq r < d$ such that $a = dq + r$.

$$q = a \operatorname{div} d \quad (\text{quotient}), \quad r = a \operatorname{mod} d \quad (\text{remainder, never negative})$$

a divisible by $d \iff r = 0$. *Example:* $-11 \operatorname{div} 3 = -4$, $-11 \operatorname{mod} 3 = 1$ (since $-11 = 3(-4) + 1$, **not** $3(-3) - 2$, since remainder can't be negative).

8.3 Congruences

Definition: for integers a, b , positive integer m : $a \equiv b \pmod{m}$ iff $m \mid (a - b)$. m is the **modulus**.

- **Theorem 5:** $a \equiv b \pmod{m} \iff \exists k \in \mathbb{Z} : a = b + km$.
- **Theorem:** $a \equiv b \pmod{m} \iff a \operatorname{mod} m = b \operatorname{mod} m$ (same remainder).
- **Theorem 6** (arithmetic with congruences): if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.
- *Fast modular exponentiation example* (repeated squaring, using Thm 6): to find $4^{64} \operatorname{mod} 13$: $4^2 \equiv 3$, $4^4 \equiv 3^2 = 9 \equiv -4$, $4^8 \equiv (-4)^2 = 16 \equiv 3$, $4^{16} \equiv 3^2 \equiv -4$, $4^{32} \equiv 3$, $4^{64} \equiv -4 \equiv 9 \pmod{13}$.

8.4 Arithmetic Modulo m ($\mathbb{Z}_m$)

$\mathbb{Z}_m = \{0, 1, \dots, m-1\}$. Define $a +_m b = (a + b) \operatorname{mod} m$, $a \times_m b = (a \times b) \operatorname{mod} m$. *Example:* in $\mathbb{Z}_{11}$: $8 +_{11} 9 = 17 \operatorname{mod} 11 = 6$; $7 \times_{11} 9 = 63 \operatorname{mod} 11 = 8$.

8.5 Primes

- $p > 1$ is **prime** if its only positive divisors are 1 and p .
- **Fundamental Theorem of Arithmetic:** every integer > 1 can be written uniquely as a prime or product of primes (in non-decreasing order).
- **Theorem:** there are infinitely many primes (Euclid's proof: assume finitely many $p_1, \dots, p_n$; $p_1 p_2 \cdots p_n + 1$ is either prime itself or has a prime factor not in the list – contradiction).

8.6 GCD and LCM

- **gcd**(a, b): largest d with $d \mid a$ and $d \mid b$.
- a, b **relatively prime** iff $\operatorname{gcd}(a, b) = 1$. **Pairwise relatively prime:** $\operatorname{gcd}(a_i, a_j) = 1$ for all $i \neq j$.
- **lcm**(a, b): smallest positive integer divisible by both a, b .
- **Via prime factorization:** gcd = product of **lowest** powers of common primes; lcm = product of **highest** powers of all primes involved.

- **Theorem 10:** $ab = \gcd(a, b) \cdot \text{lcm}(a, b)$.
- *Example:* $150 = 2 \cdot 3 \cdot 5^2$, $500 = 2^2 \cdot 5^3 \Rightarrow \gcd = 2 \cdot 5^2 = 50$, $\text{lcm} = 2^2 \cdot 3 \cdot 5^3 = 1500$ (check: $50 \times 1500 = 150 \times 500$).

8.7 The Euclidean Algorithm

Lemma: if $a = bq + r$ then $\gcd(a, b) = \gcd(b, r)$. Repeatedly apply division algorithm until remainder 0; last nonzero remainder is the gcd.

Worked example: $\gcd(2415, 945)$:

$$\begin{aligned} 2415 &= 945 \times 2 + 525 \\ 945 &= 525 \times 1 + 420 \\ 525 &= 420 \times 1 + 105 \\ 420 &= 105 \times 4 + 0 \quad \Rightarrow \quad \gcd(2415, 945) = 105 \end{aligned}$$

8.8 Bézout's Theorem (gcd as Linear Combination)

Theorem: if a, b positive integers, $\exists s, t \in \mathbb{Z} : \gcd(a, b) = sa + tb$. Find s, t by **back-substitution** through the Euclidean algorithm steps.

Worked example: express $\gcd(252, 198) = 18$ as $s(252) + t(198)$.

$$\begin{aligned} 252 &= 1 \times 198 + 54 \\ 198 &= 3 \times 54 + 36 \\ 54 &= 1 \times 36 + 18 \\ 36 &= 2 \times 18 \quad \Rightarrow \quad \gcd = 18 \end{aligned}$$

Back-substitute:

$$\begin{aligned} 18 &= 54 - 1 \times 36 \\ &= 54 - 1(198 - 3 \times 54) = 4 \times 54 - 1 \times 198 \\ &= 4(252 - 1 \times 198) - 1 \times 198 = 4 \times 252 - 5 \times 198 \end{aligned}$$

So $\gcd(252, 198) = 18 = 4(252) - 5(198)$.

Applications: **Lemma:** if $\gcd(a, b) = 1$ and $a \mid bc$ then $a \mid c$. **Lemma:** if p prime and $p \mid a_1 a_2 \cdots a_n$, then $p \mid a_i$ for some i . **Theorem 15:** if $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$ (cancellation law – **only** valid when $\gcd(c, m) = 1$).

8.9 Solving Linear Congruences

$ax \equiv b \pmod{m}$: find all integers x satisfying this.

- a' is an **inverse of a modulo m** if $a'a \equiv 1 \pmod{m}$.
- **Theorem 16:** if $\gcd(a, m) = 1$ ($m \geq 2$), a unique inverse $\bar{a}$ exists with $0 < \bar{a} < m$.
- **To find the inverse:** use Extended Euclidean Algorithm (Bézout) to write $sa + tm = 1$; then s (reduced mod m) is the inverse of a .
- **To solve $ax \equiv b \pmod{m}$:** find inverse $\bar{a}$ of a ; multiply both sides: $x \equiv \bar{a}b \pmod{m}$.

Worked example: solve $3x \equiv 4 \pmod{7}$. $\gcd(3, 7) = 1$; find inverse of 3 mod 7: since $3 \times 5 = 15 \equiv 1 \pmod{7}$, inverse is 5. Then $x \equiv 5 \times 4 = 20 \equiv 6 \pmod{7}$. So $x \equiv 6 \pmod{7}$ (i.e. $x = 6, 13, 20, \dots$ or $x = \dots, -1, 6, 13, \dots$).

8.10 Euler's Totient Function and Euler's Theorem

- $\varphi(n)$ = number of integers in $\{1, \dots, n\}$ relatively prime to n (**Euler's totient function**).
 $\varphi(p) = p - 1$ for prime p .
- **Reduced residue system mod n** : a set of $\varphi(n)$ integers, one from each residue class relatively prime to n .
- **Theorem 20 (Euler)**: if $n > 0$, $\gcd(a, n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.
- **Fermat's Little Theorem** (special case, $n = p$ prime): if $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.